

Auctions

1. In a second-price (Vickrey) auction, the winner ...
 - (a) pays the average of all bids
 - (b) is the highest bidder and pays the second-highest bid
 - (c) is chosen at random
 - (d) is the highest bidder and pays their own bid
2. In a second-price auction, bidding your true value is ...
 - (a) the same as bidding zero
 - (b) a weakly dominant strategy
 - (c) never optimal
 - (d) only optimal if you know the others' values
3. A Bayesian Nash equilibrium is a strategy profile in which ...
 - (a) the seller chooses the winner
 - (b) all players bid the same amount
 - (c) each player maximises expected utility given their type and correct beliefs about others
 - (d) every player knows everyone else's value
4. In a first-price auction, a rational bidder usually ...
 - (a) shades their bid below their true value
 - (b) always bids zero
 - (c) bids their true value
 - (d) bids above their true value
5. The revenue equivalence theorem says that ...
 - (a) first-price and second-price auctions raise the same expected revenue (under its conditions)
 - (b) the winner always pays the same as the loser
 - (c) the seller always prefers a first-price auction
 - (d) revenue is always zero
6. Because truthful bidding is dominant, the second-price auction is ...
 - (a) always more profitable for the seller
 - (b) impossible to run with sealed bids
 - (c) only suitable for two bidders

- (d) efficient: the highest-value bidder wins
7. An auction game is specified by ...
 - (a) a single payoff matrix
 - (b) a discount factor
 - (c) the bidders' value distributions, an allocation rule and a payment rule
 - (d) a list of stable matchings
 8. In a first-price auction, the winner pays ...
 - (a) nothing
 - (b) the average bid
 - (c) their own bid
 - (d) the second-highest bid
 9. A bidder's type in an auction is ...
 - (a) their private value for the item
 - (b) their bid
 - (c) the reserve price
 - (d) the number of other bidders
 10. In a first-price auction with N bidders whose values are uniform on $[0, 1]$, the equilibrium bid is ...
 - (a) $v/2$ for all N
 - (b) $\frac{N-1}{N}v$
 - (c) v
 - (d) 0
 11. A bidder's utility in an auction is ...
 - (a) always zero
 - (b) $v_i q_i - p_i$: value times allocation, minus payment
 - (c) their bid
 - (d) the seller's revenue
 12. Revenue equivalence requires that the auction ...
 - (a) always gives the item to the highest-value bidder and gives a zero-value bidder zero surplus
 - (b) has exactly two bidders
 - (c) charges a reserve price

- (d) uses sealed bids
13. In a second-price auction the winner's surplus is ...
- (a) the second-highest value
 - (b) zero
 - (c) their value minus their own bid
 - (d) their value minus the second-highest value
14. Truthful bidding being a dominant strategy means that in a second-price auction ...
- (a) impossible to run sealed-bid
 - (b) bidders need not guess others' values to bid optimally
 - (c) only suitable for one bidder
 - (d) more profitable than any other auction
15. The second-price auction is efficient because ...
- (a) the seller earns the most
 - (b) no one ever wins
 - (c) everyone pays the same
 - (d) the bidder who values the item most wins it
16. For N values drawn independently and uniformly on $[0, 1]$, the expected maximum is ...
- (a) $\frac{1}{N}$
 - (b) $\frac{1}{2}$
 - (c) $\frac{N-1}{N+1}$
 - (d) $\frac{N}{N+1}$
17. In a first-price auction with N uniform bidders bidding $\frac{N-1}{N}v$, the seller's expected revenue is ...
- (a) $\frac{N-1}{N+1}$
 - (b) $\frac{N}{N+1}$
 - (c) $\frac{1}{N}$
 - (d) $\frac{1}{2}$
18. In a second-price auction with N uniform bidders, the seller's expected revenue equals ...
- (a) the expected second-highest value, $\frac{N-1}{N+1}$
 - (b) the expected highest value, $\frac{N}{N+1}$
 - (c) $\frac{1}{2}$

- (d) zero
19. A bidding strategy in an auction game is a map $b_i : V_i \rightarrow B_i$ that sends ...
- (a) a bid to a payment
 - (b) a valuation to a payment
 - (c) a bidder's valuation to a bid
 - (d) a bid to an allocation
20. The allocation rule q of an auction maps the profile of bids to ...
- (a) the probability with which each bidder receives the good
 - (b) the amount each bidder pays
 - (c) each bidder's private valuation
 - (d) the seller's reserve price